

Chapter 23 Wave-like properties of matter

23.1 De Broglie's Hypothesis

photon:



dual particle-
wave nature

$$E = h\nu,$$

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda} = \frac{h}{2\pi} \frac{2\pi}{\lambda} = \hbar k.$$

$$\mathbf{p} = \hbar \mathbf{k}.$$

k —wave number (number of
waves in a length of 2π unit)

$\mathbf{k}$ —wave vector

In 1923, Louis Victor de Broglie suggested that associated with any material particle moving with momentum p , there is a wave of wavelength

$$\lambda = \frac{h}{p} \quad (\text{de Broglie wavelength}).$$

with no experimental evidence at all

His idea is based on

- The relativistic transformation relations can be shared by both physical quantities E , $\mathbf{p}$ of the particle and ω , $\mathbf{k}$ of the wave.

Four dimensional vectors:

$$x_\mu = (\mathbf{r}, ict),$$

$$A'_\mu = a_{\mu\nu} A_\nu.$$

$$p_\mu = (\mathbf{p}, \frac{i}{c} E),$$

$$E = h\nu = \hbar\omega,$$

$$k_\mu = (\mathbf{k}, \frac{i}{c} \omega),$$

$$p = \frac{h}{\lambda} = \hbar k, \quad \mathbf{p} = \hbar \mathbf{k}.$$

...

If the relative motion is in the x-direction, we have

$$a = \begin{pmatrix} \gamma & 0 & 0 & i\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\beta\gamma & 0 & 0 & \gamma \end{pmatrix}$$

- Bohr's quantization condition for the stationary state can be neatly interrupted in a standing wave picture.

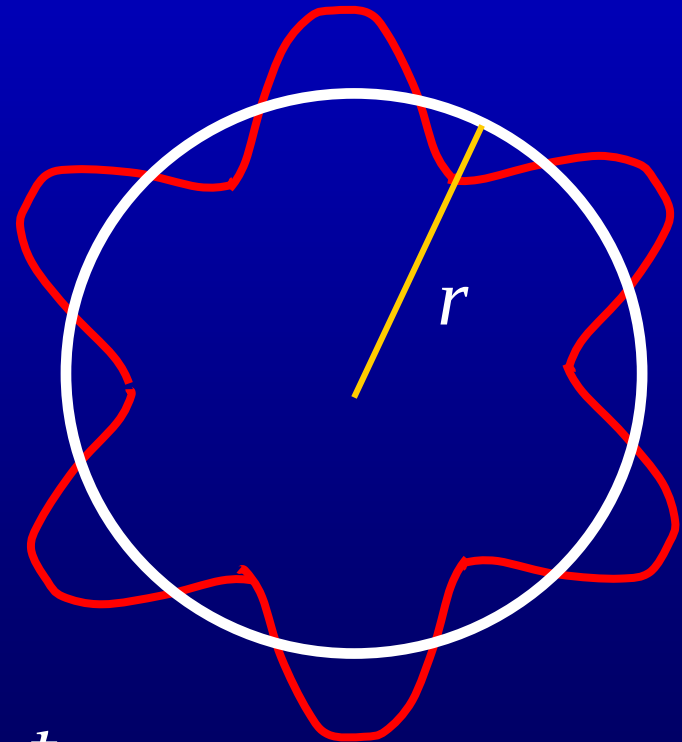
condition for standing wave:

$$2\pi r = n\lambda.$$

de Broglie's relation:

$$\lambda = \frac{h}{p}.$$

$$2\pi r = n \frac{h}{p} \Rightarrow pr = mvr = n \frac{h}{2\pi} = n\hbar.$$

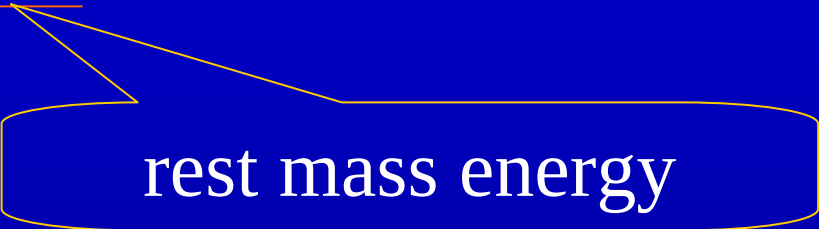




Prince Louis Victor de Broglie
1892-1987
1929 Nobel Prize in Physics

The relativistic kinetic energy can be written as

$$E_K = mc^2 - \underline{m_0c^2}.$$



rest mass energy

From energy momentum relation

$$E^2 = m_0^2c^4 + p^2c^2$$

we have

$$\begin{aligned} p &= \frac{1}{c} \sqrt{E^2 - m_0^2c^4} = \frac{1}{c} \sqrt{(E - m_0c^2)(E + m_0c^2)} \\ &= \frac{1}{c} \sqrt{E_K (E_K + 2m_0c^2)}. \end{aligned}$$

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{E_K(E_K + 2m_0c^2)}}$$

$$= \frac{h}{\sqrt{2m_0E_K \left[1 + \frac{E_K}{2m_0c^2} \right]}}.$$

When $E_K \ll m_0c^2$, $E_K \approx \frac{1}{2}m_0v^2$,

$$\lambda \approx \frac{h}{\sqrt{2m_0E_K}} \approx \frac{h}{m_0v}.$$

At extremely relativistic limit, $E_K \gg m_0 c^2$, we have

$$\lambda \approx \frac{hc}{E_K} = \frac{1240 \text{ eV} \cdot \text{nm}}{E_K}.$$

For a non - relativistic electron, $E_K \ll m_0 c^2$, we have

$$\lambda \approx \frac{h}{\sqrt{2m_0 E_K}} = \sqrt{\frac{150.4 \text{ eV}}{E_K}} \text{ \AA}.$$

Example Compute the de Broglie wavelength of following:

- (1) A 1000-kg automobile travelling at 100m/s.
- (2) A 10g bullet travelling at 500m/s.
- (3) A smoke particle of mass 10^{-6} g moving at 1cm/s.
- (4) An electron with kinetic energy of 1eV.

Solution: (1)

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J}\cdot\text{s}}{10^3 \text{ kg} \times 100 \text{ m/s}} = 6.6 \times 10^{-39} \text{ m}.$$

(2)

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J}\cdot\text{s}}{10^{-2} \text{ kg} \times 500 \text{ m/s}} = 1.3 \times 10^{-34} \text{ m}.$$

(3)

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J}\cdot\text{s}}{10^{-9} \text{ kg} \times 10^{-2} \text{ m/s}} = 6.6 \times 10^{-23} \text{ m}.$$

(4)

$$\begin{aligned} \lambda &= \frac{h}{\sqrt{2m_e E_K}} = \frac{6.6 \times 10^{-34} \text{ J}\cdot\text{s}}{\sqrt{2 \times 9.1 \times 10^{-31} \text{ kg} \times 1.6 \times 10^{-19} \text{ J}}} \\ &= 1.2 \times 10^{-9} \text{ m} = 1.2 \text{ nm}. \end{aligned}$$

Only for particles of atomic or nuclear size the wave behavior is observable.

23.2 Davisson-Germer experiment

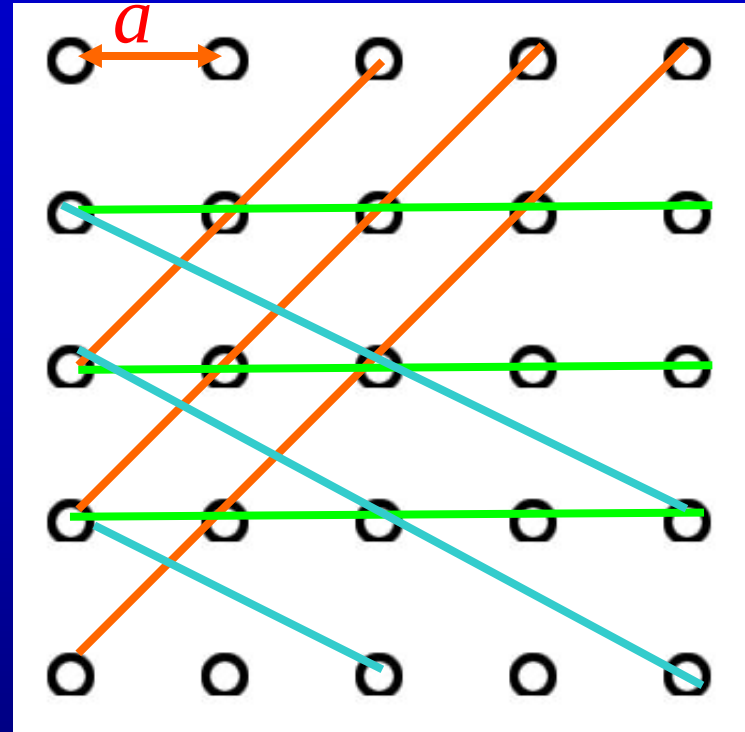
a -lattice constant

families of parallel planes

——crystal planes

X rays: $\lambda \sim 0.1 - 100 \text{ \AA}$

$a \sim 1 \text{ \AA}$.



The condition for constructive interference is

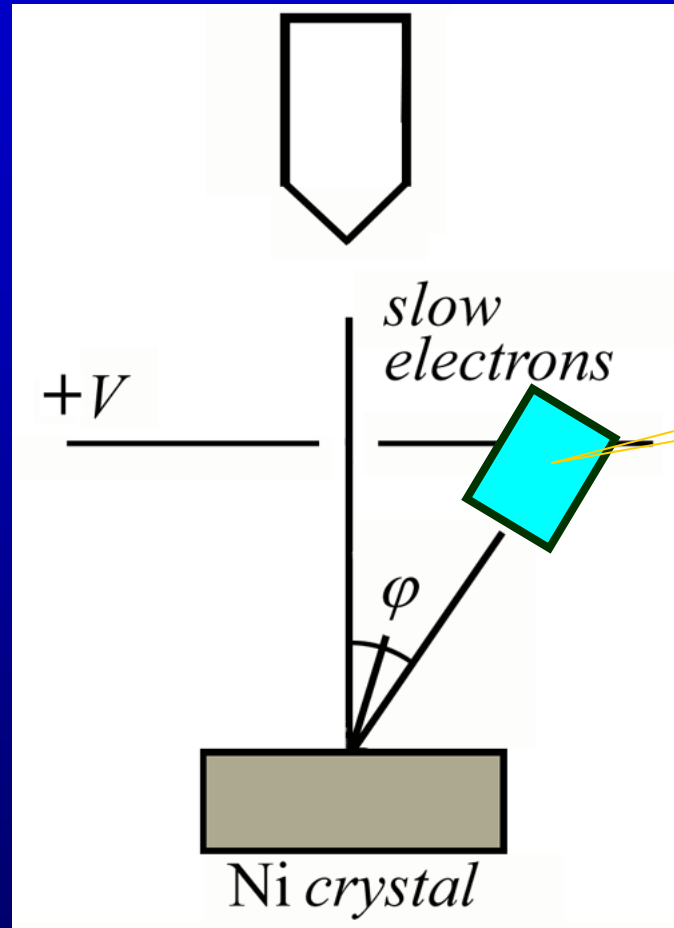
$$2d \sin \theta = n\lambda, \quad n = 1, 2, 3, \dots$$

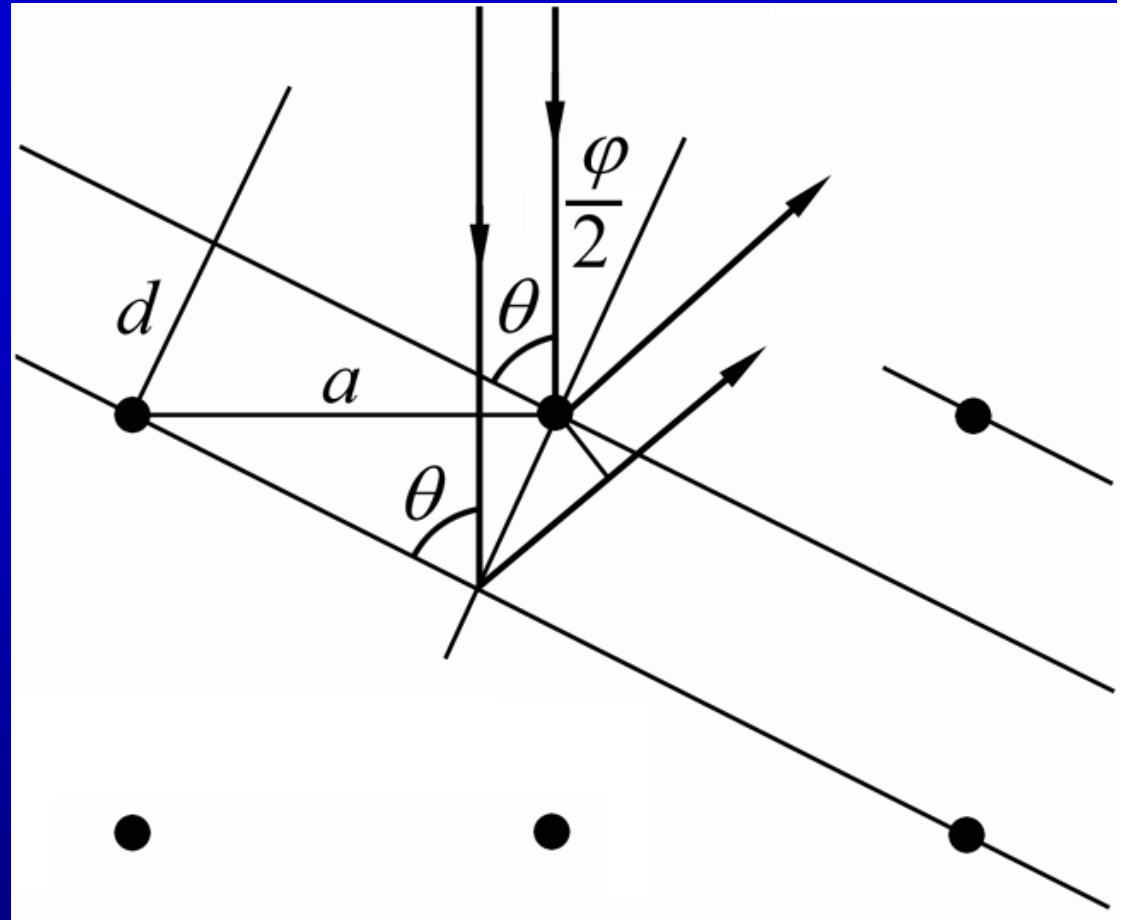
Bragg's condition

In 1923 Clinton Davisson published some work on the angular distribution of electrons reflected from a sheet of platinum

Born suggested that this might be an interference phenomenon related to de Broglie's idea on the connection of plane waves with quantum particles.

Davisson and Lester Germer (1926)



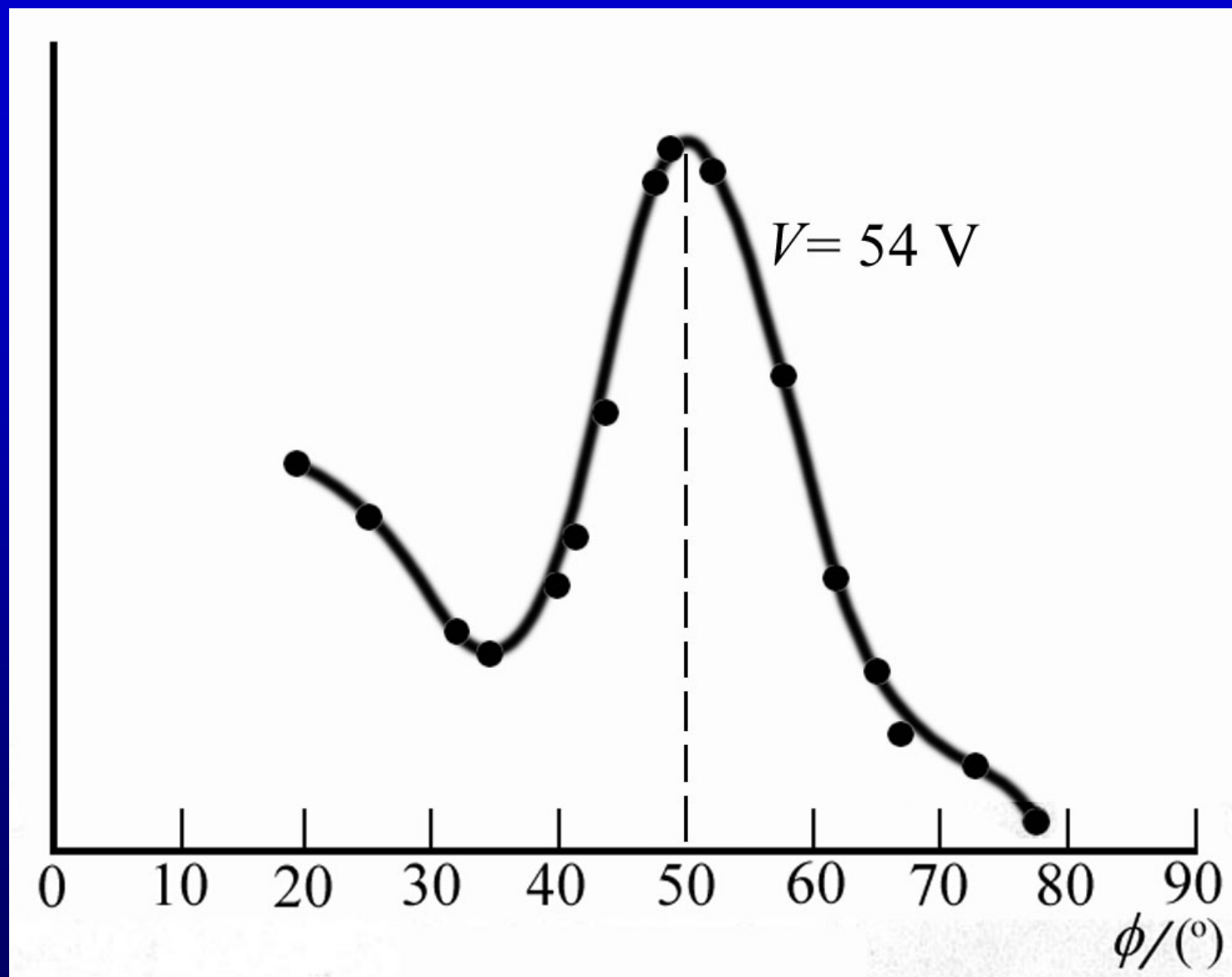


$$\theta = 90^\circ - \varphi / 2$$

$$d = a \sin (\varphi / 2)$$

$$2d \sin \theta = n\lambda$$

$$2d \sin \theta = 2a \sin(\varphi/2) \cos(\varphi/2) = a \sin \varphi$$



$$a \sin \varphi = n\lambda \quad (n = 1, 2, \dots)$$

lattice constant = 0.215nm

$V = 54\text{eV}$, $\varphi = 50^\circ$, there is a maximum
 $\lambda = 0.165\text{nm}$

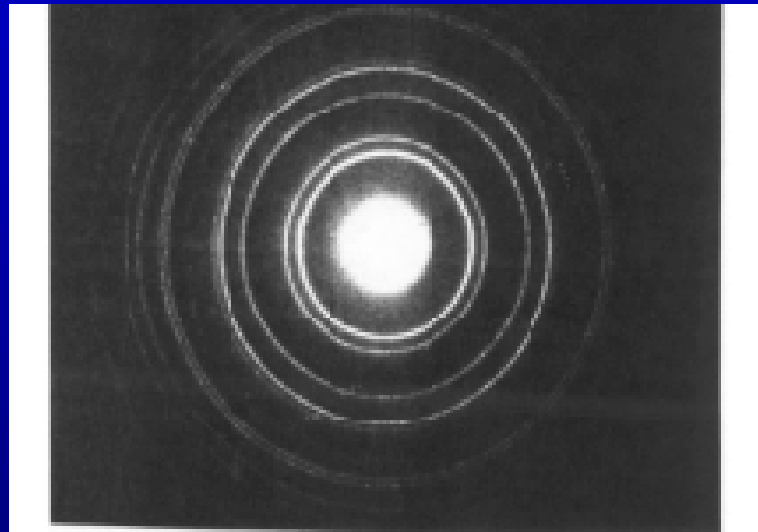
$$V = 54\text{eV},$$

$$\begin{array}{c} \downarrow \\ \rightarrow p \xrightarrow{h} \lambda_{\text{de Broglie}} = 0.167\text{nm} \end{array}$$

$$\lambda = 0.165\text{nm} \longrightarrow h \sim 1\%$$

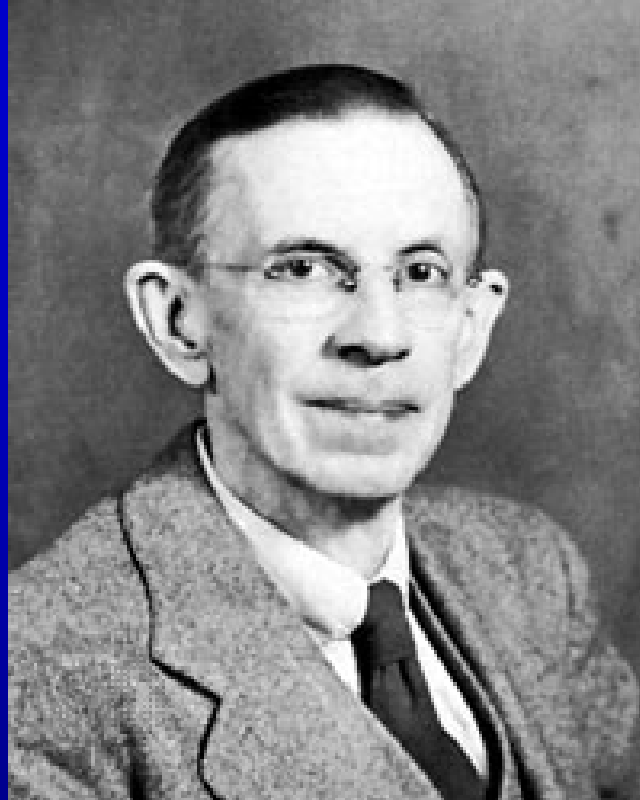
G. P. Thomson (1927) used beam of 600eV electrons to hit aluminium foil crystallite

$\lambda = 0.5\text{\AA}$ the Debye pattern



While J. J. Thomson found particle electron, ...

Davisson and Thomson shared the 1937 Nobel Prize.



Clinton Joseph Davisson



Joseph John Thomson

1856-1940

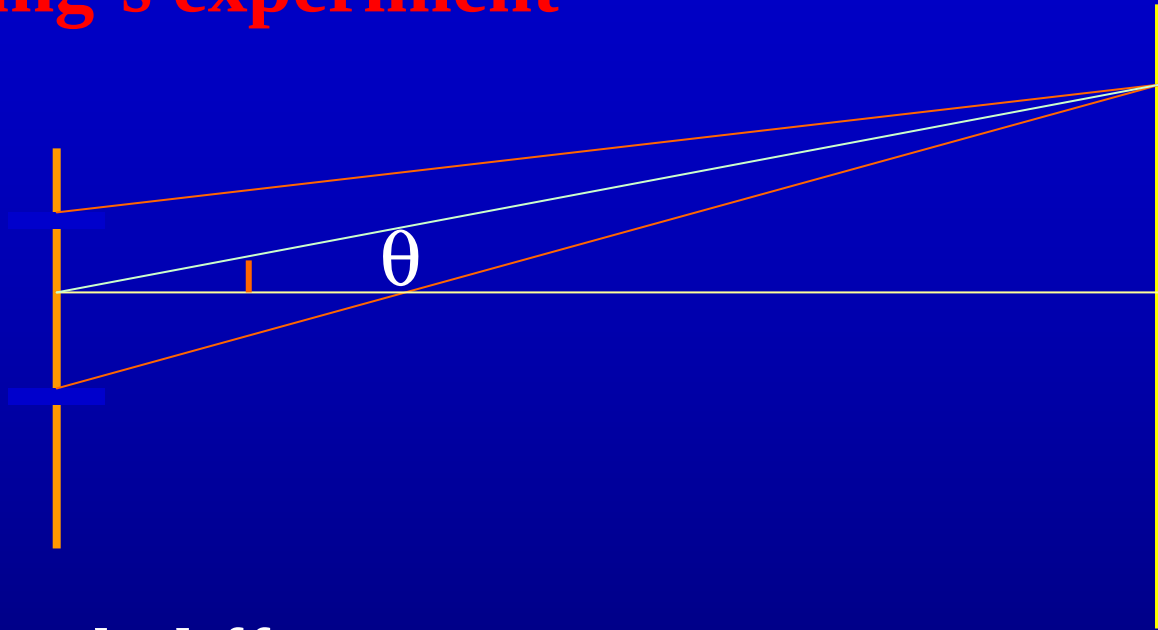


George Paget Thomson

1892-1975

23.3 Electron double-slit experiment

Young's experiment

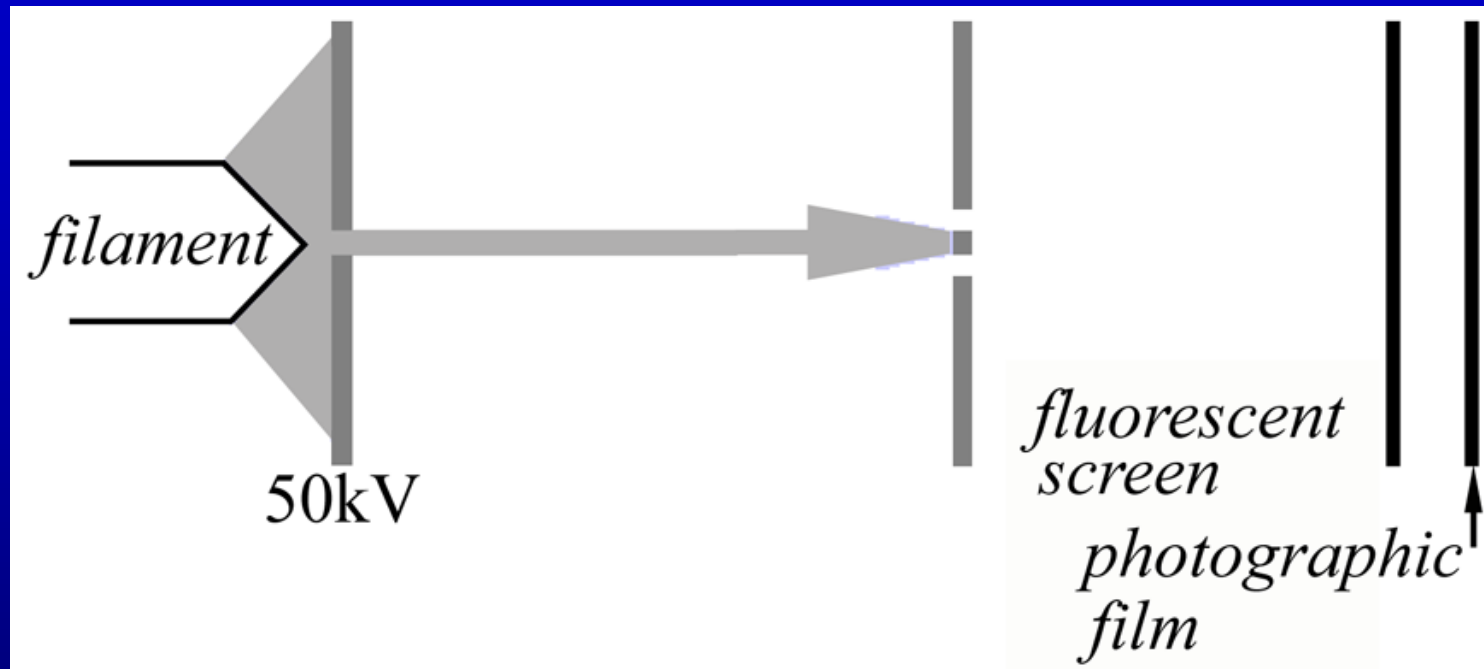


The path difference is $d \sin \theta$.

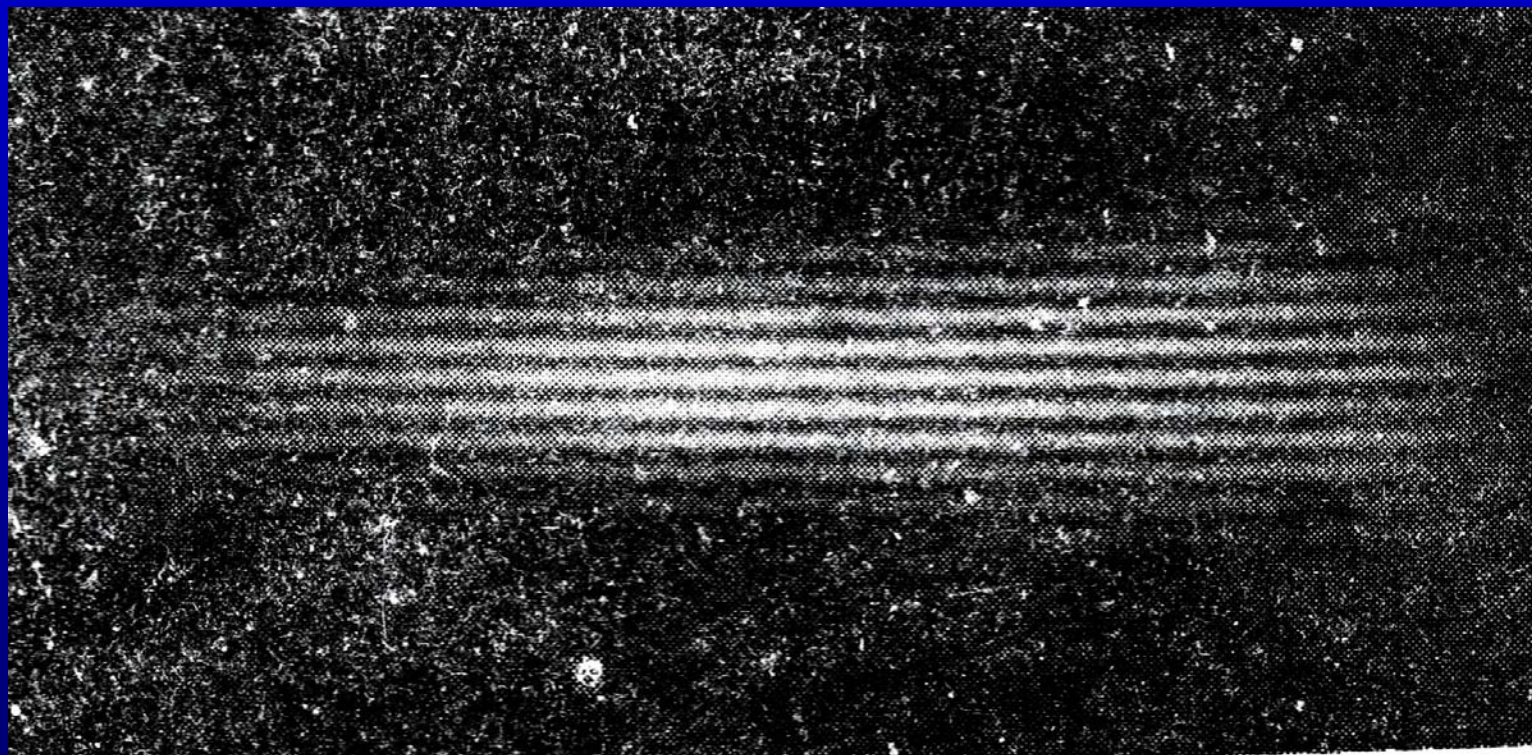
$$d \sin \theta = m\lambda \quad \text{constructive interference}$$

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda \quad \text{destructive interference}$$

1961 Claus Jönsson has done the experiment with electrons.



Its similarity with pattern of light is striking and shows evidence for the wave nature of electrons.



Even the electron shot one by one, the interference pattern is maintained.

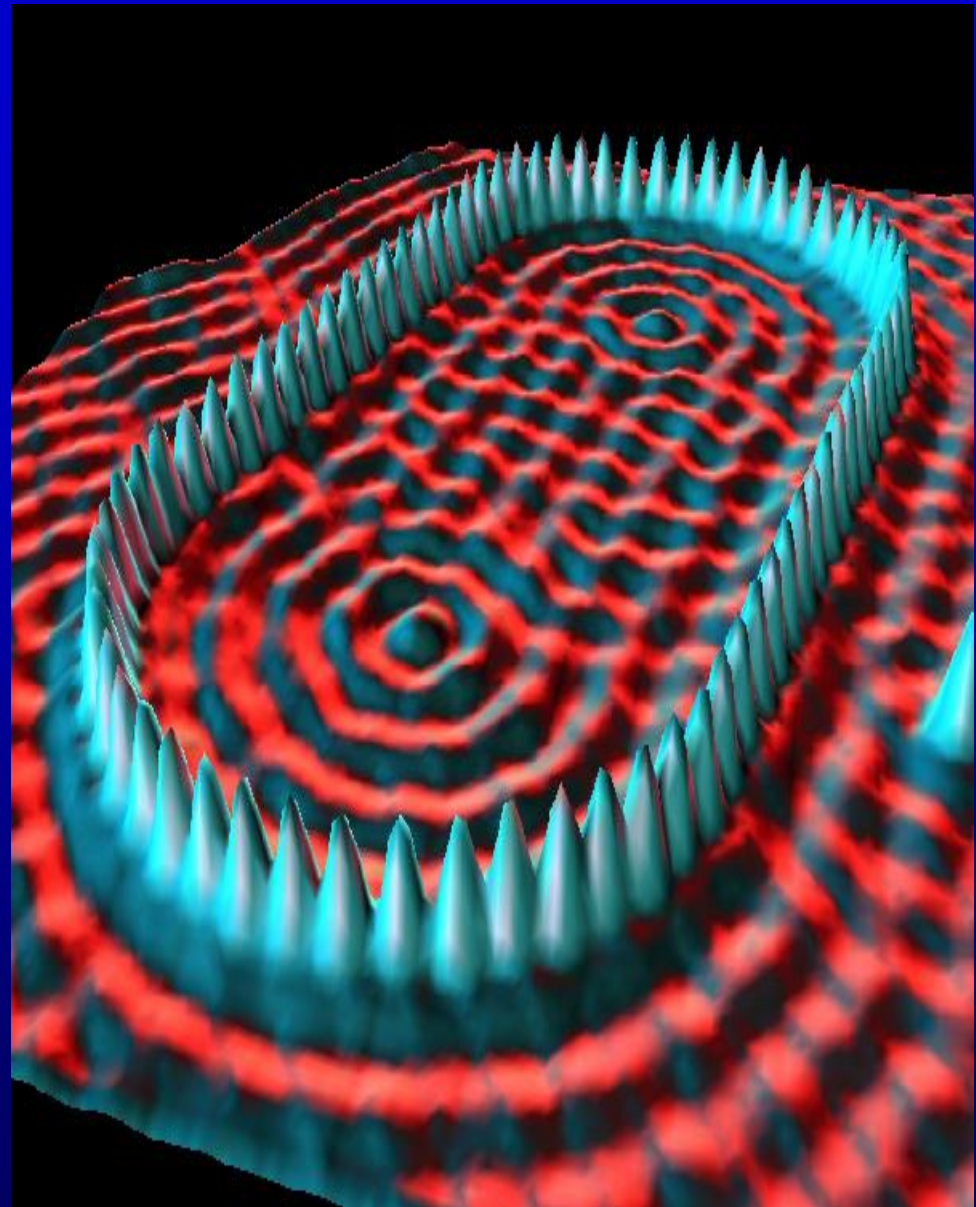
But no matter what means we try in order to determine through which slit an electron passes, we find that the interference pattern is destroyed — any effort to investigate the particle aspect will destroy the pattern—wave aspect.

Bohr's Principle of complementarity

The complete description of a physical entity such as photon cannot be done in terms of particles or waves exclusively, but both aspects must be considered. Moreover, the particle and wave natures cannot be observed simultaneously.

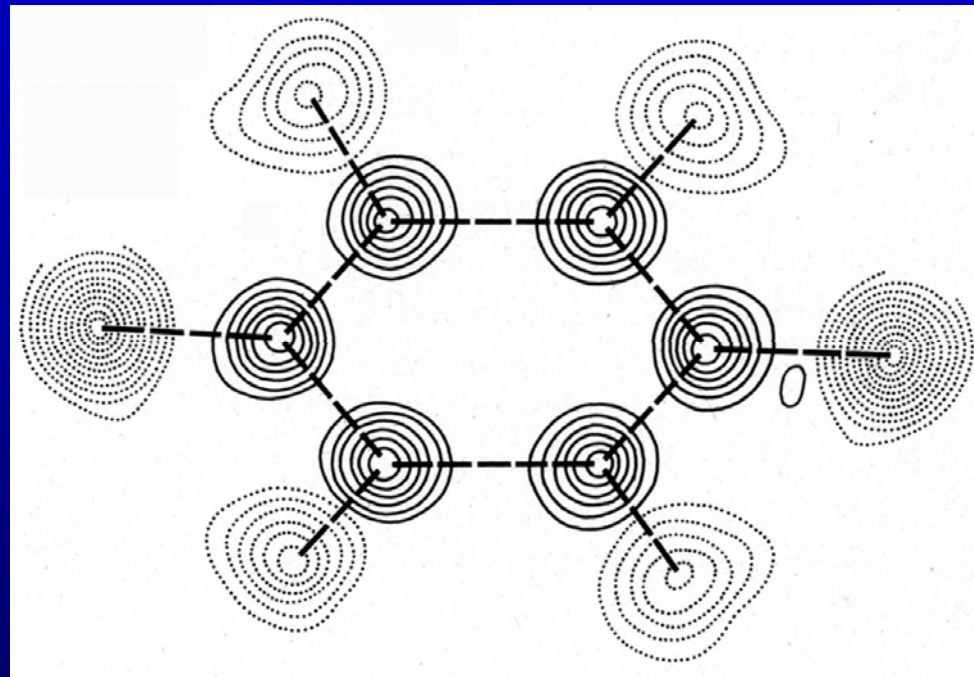
Quantum Corral

STM picture of a stadium-shaped "quantum corral" made by positioning iron atoms on a copper surface. Surface electron waves in a confined region. Courtesy, Don Eigler, IBM



Wave property of matter particle is used extensively as probe.

Neutron diffraction



benzene

0.1nm neutron

23.4 Uncertainty relation

To measure wavelength of a wave

$$\lambda = \frac{\Delta x}{n}$$

The deviation of n is smaller than or equal to ± 1

$$\Delta\lambda \leq \frac{\Delta x}{n} - \frac{\Delta x}{n+1} = \frac{\lambda}{n+1}$$

$$\Delta x \Delta\lambda \sim n\lambda \frac{\lambda}{n+1} \sim \lambda^2$$

$$\Delta x \frac{\Delta\lambda}{\lambda^2} \sim \Delta x \Delta\left(\frac{1}{\lambda}\right) \sim 1$$

As to frequency, we count number of oscillations n in interval Δt .

$$\nu = \frac{n}{\Delta t}$$

$$\Delta \nu = \frac{n+1}{\Delta t} - \frac{n}{\Delta t}$$

$$\Delta \nu \cdot \Delta t \sim 1$$

For wave packets (see Appendix E) of different dispersion,

$$\Delta k \Delta x \geq 1$$

Werner Curt Heisengberg (1901—1976) asserted that position and momentum of a particle could not be simultaneously measured.

$$\Delta x \cdot \Delta p \geq \hbar$$

More precisely

$$\Delta x \cdot \Delta p \geq \frac{1}{2} \hbar$$

For matter particle, a wavelength $\lambda = h / p$

$$p = \frac{h}{\lambda} = \hbar \frac{2\pi}{\lambda} \equiv \hbar k$$

$$\Delta x \cdot \Delta p = \Delta x \cdot \Delta(\hbar k) \geq \hbar \longleftrightarrow \Delta x \cdot \Delta k \geq 1$$

nothing but a wavelike property

Example 23.3 Single slit diffraction.

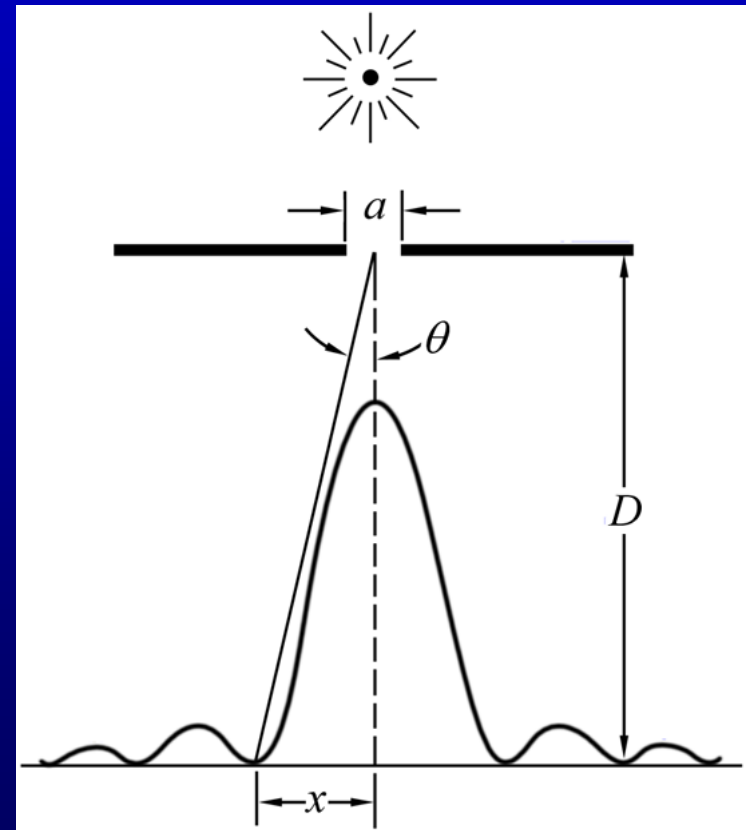
Initially

$$p_x = 0, \Delta p_x = 0,$$

$$\Delta x = \infty.$$

When the particle passes through the slit

$$\Delta x \sim a, \rightarrow \Delta p_x \sim \frac{\hbar}{a}.$$



For first minimum

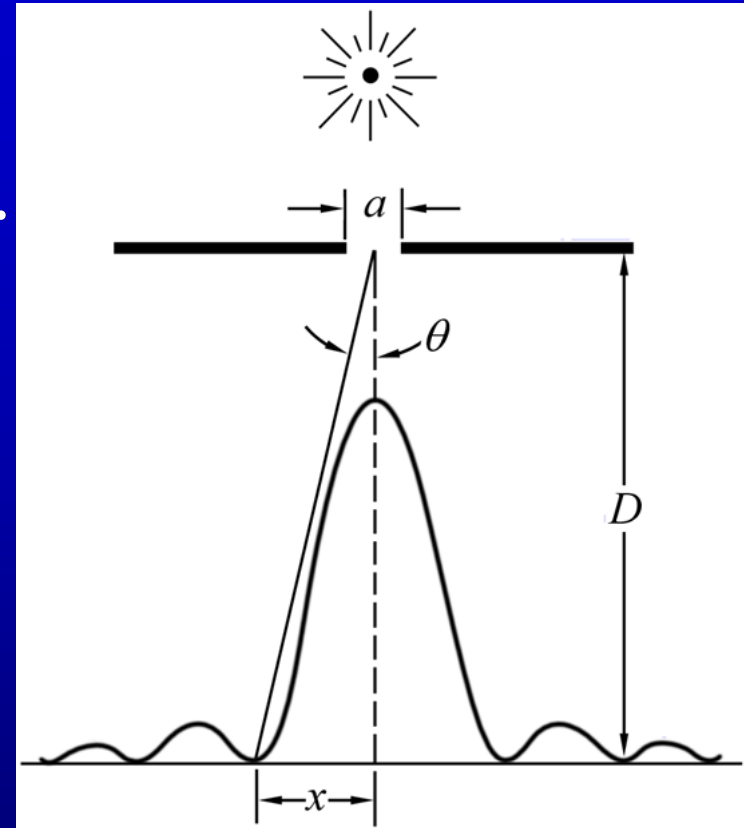
$$a \sin \theta = \lambda, \quad \sin \theta \approx \tan \theta = \frac{x}{D}.$$

$$\lambda = \frac{h}{p_0}.$$

$$a \frac{x}{D} = \frac{h}{p_0}.$$

$$\frac{p_x}{p_0} = \frac{x}{D}.$$

$$p_x = p_0 \frac{x}{D} = \frac{h}{a}$$



Example 23.4 Beta-decay

For a nucleus of several fms, the uncertainty of position is

$$\Delta x = 10^{-14} \text{ m}$$

According to uncertainty principle, the uncertainty in momentum is

$$\Delta p \sim \frac{\hbar}{10^{-14} \text{ m}} \sim 10^{-20} \text{ kg} \frac{\text{m}}{\text{s}}$$

$$E_k = \sqrt{c^2 p^2 + m_0^2 c^4} - m_0 c^2 \sim 20 \text{ MeV}$$

Usually beta-decay kinetic energy is smaller than 1MeV.....

Example Melons in Quantum Land

In a strange quantum land, $\hbar = 10^{-3} \text{ J} \cdot \text{s}$ Melons with very hard peels are grown. The diameter of a melon is about 20cm and the mass of a seed is about 0.1g.

Why do we have to be careful when cut open the melon?

Solution: From the uncertainty relation $\Delta x \cdot \Delta p \approx \hbar$.

The momentum uncertainty of the melon seeds is

$$\Delta p \approx \frac{10^{-3} \text{ J} \cdot \text{s}}{0.1 \text{ m}} = 10^{-2} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}.$$

The corresponding velocity uncertainty is

$$\Delta v = \frac{\Delta p}{m} \approx 10^2 \text{ m} \cdot \text{s}^{-1}.$$

Example 23.6 Simple harmonic oscillator

$$E = \frac{p^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2$$

If we replace p by x through uncertainty relation

$$E = \frac{1}{2\mu} \frac{\hbar^2}{4x^2} + \frac{1}{2}\mu\omega^2 x^2.$$

Then

$$E \geq 2\sqrt{\frac{1}{2\mu} \frac{\hbar^2}{4x^2} \cdot \frac{1}{2}\mu\omega^2 x^2} = \frac{\hbar\omega}{2}.$$

$$E_{\min} = \frac{\hbar\omega}{2}, x^2 = \frac{\hbar^2}{2\mu\omega}.$$

In measurement process, uncertainty means

$$\Delta x \equiv \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

Uncertainty relation comes from wave nature of matter particles.

Werner Heisenberg won 1932 Nobel Prize “for the creation of quantum mechanics”.

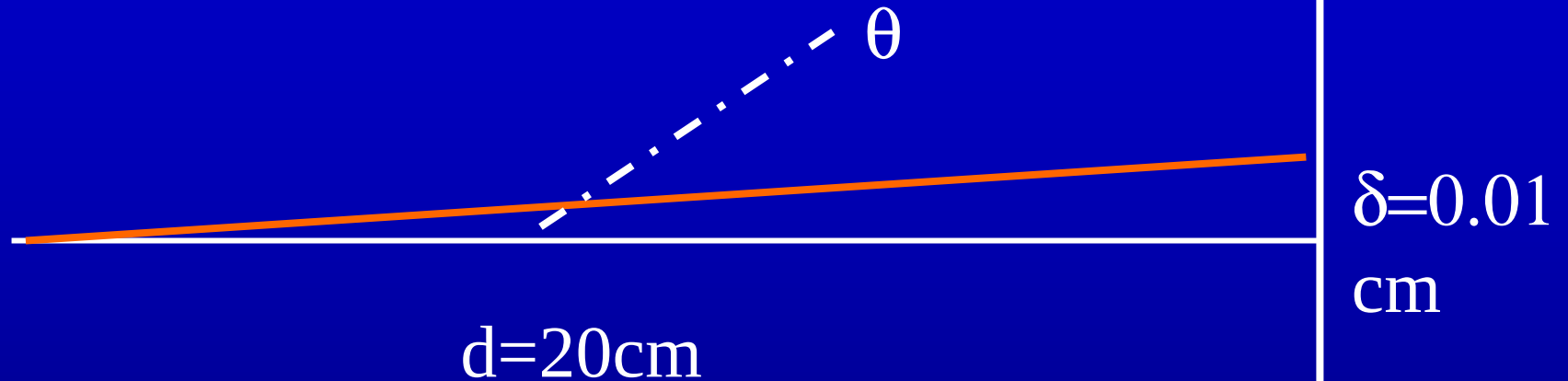
Erwin Schrödinger and Paul Adrien Maurice Dirac share the 1933 Nobel Prize “for the discovery of new productive forms of atomic theory”.

1. A dust particle of $10^{-12} g$

If $\Delta x \approx 10^{-4} cm$, then $\Delta v \approx \frac{\hbar}{m\Delta x} \approx 5 \times 10^{-13} m/s$.

2. An electron moving in a TV-tube

$$K = 3000eV \quad p = 3 \times 10^{-23} \text{ kgm/s}$$



$$\Delta\theta = \frac{0.01}{20} = 5 \times 10^{-4} \text{ radians}$$

$$\Delta p \approx 3 \times 10^{-23} \times 5 \times 10^{-4} = 1.5 \times 10^{-26} \text{ kgm/s}$$

$$\Delta x \approx \frac{\hbar}{\Delta p} \approx 10^{-6} \text{ cm} \ll 0.01 \text{ cm}$$

3. An electron in an atom

$$v \sim 10^8 \text{ cm} / \text{s}$$

$$\Delta x \sim 10^{-8} \text{ cm}$$

$$\Delta v \approx \frac{\hbar}{2m_e \Delta x} \sim 0.6 \times 10^8 \text{ cm} / \text{s} \quad \text{comparable to } v$$

Another important pair is energy and time

$$\Delta E \cdot \Delta t \geq \frac{1}{2} \hbar.$$

An excited state has certain **lifetime**. Eq. (23.4.7) can be used to estimate the **width of an energy level**, thence the width of spectrum line.

Assignment: 23.2, 23.6, 23.8

C. Davisson and L.H. Germer, "Diffraction of electrons by a crystal of Nickel," *Physical Review* 30 (6), 705–741 (1927). You can read this paper [here](#).

THE investigation reported in this paper was begun as the result of an accident which occurred in this laboratory in April 1925. At that time we were continuing an investigation, first reported in 1921,¹ of the distribution-in-angle of electrons scattered by a target of ordinary (polycrystalline) nickel. During the course of this work a liquid-air bottle exploded at a time when the target was at a high temperature; the experimental tube was broken, and the target heavily oxidized by the intrushing air. The oxide was eventually reduced and a layer of the target removed by vaporization, but only after prolonged heating at various high temperatures in hydrogen and in vacuum.

When the experiments were continued it was found that the distribution-in-angle of the scattered electrons had been completely changed. Specimen curves exhibiting this alteration are shown in Fig. 1. These curves are all for a bombarding potential of 75 volts. The electron beam is incident on the target from the right, and the intensities of scattering in different directions are proportional to the vectors from the point of bombardment to the curves. The upper curves (for different angles of incidence) are characteristic of the target prior to the accident. They are of the type described in the note in "Science" in 1921, and are similar to curves that have been obtained for nickel in four or five other experiments. The lower curves—obtained after the accident—were the first of their sort to be observed. This marked alteration in the scattering pattern was traced to a re-crystallization of the target that occurred during the prolonged heating. Before the accident and in previous experiments we had been bombarding many small crystals, but in the tests subsequent to the accident we were bombarding only a few large ones. The actual number was of the order of ten.